

STEPS FOR FPGA IMPLEMENTATION



CREATING A PROJECT
PROGRAM & SCHEMATIC
GENERATING TESTBENCH
SYNTHESIS
IMPLEMENTATION

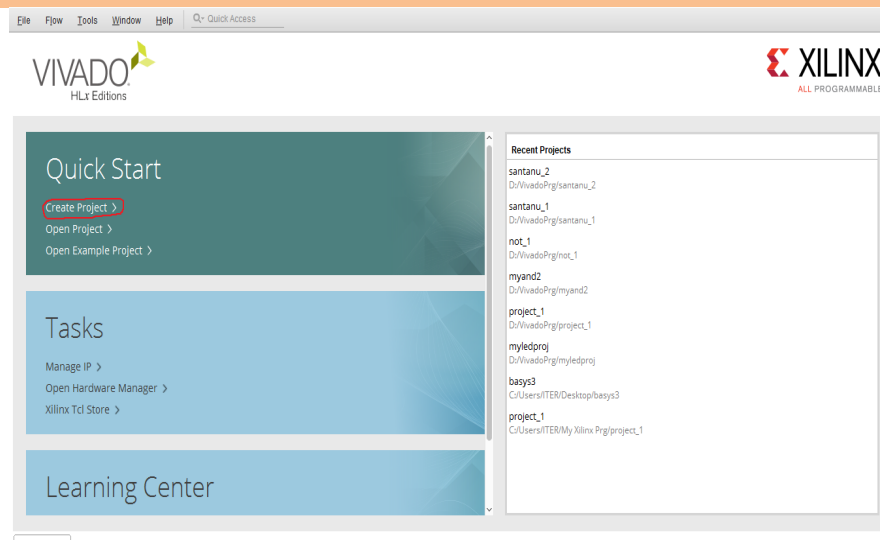
STEPS

1. CREATING A PROJECT

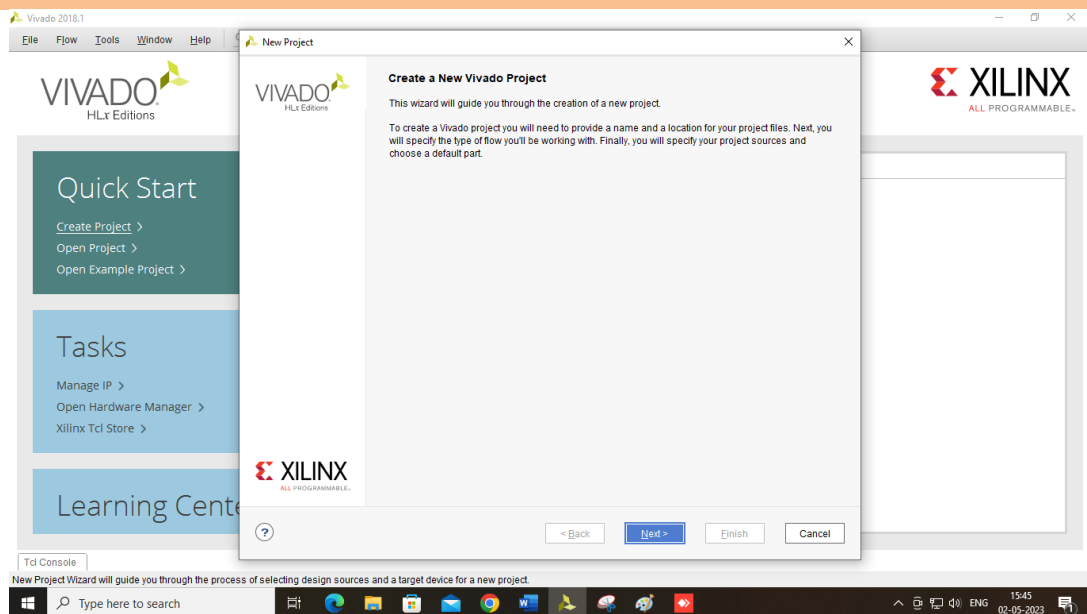
1



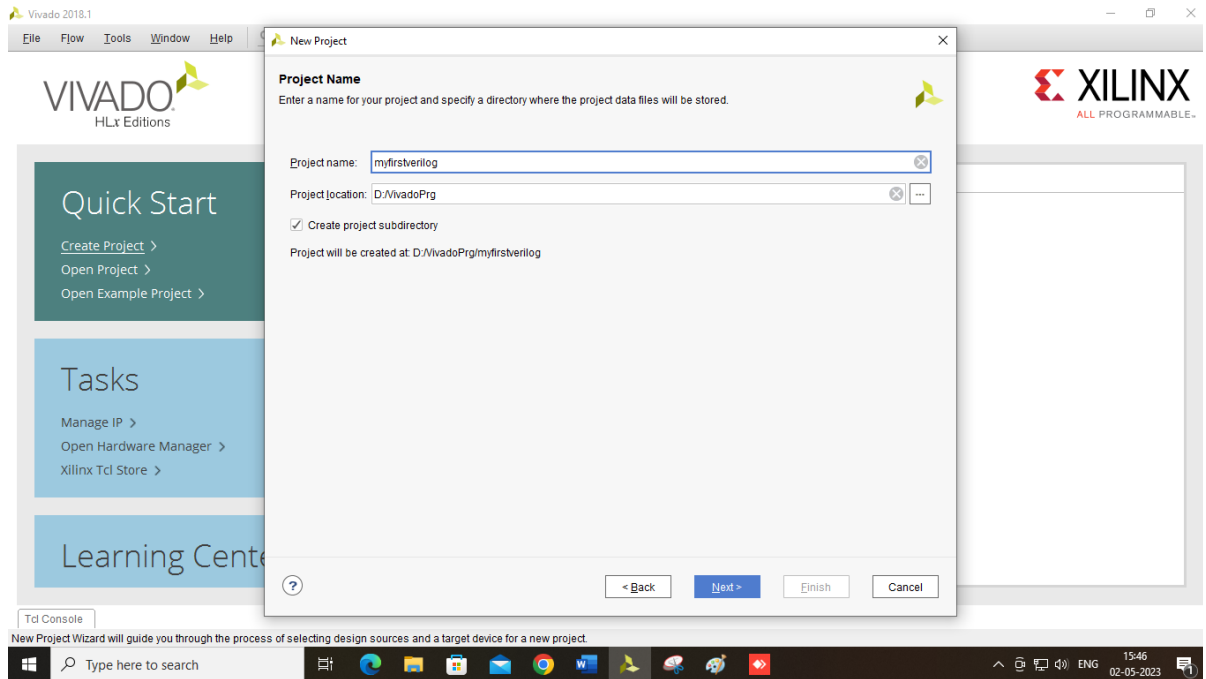
2



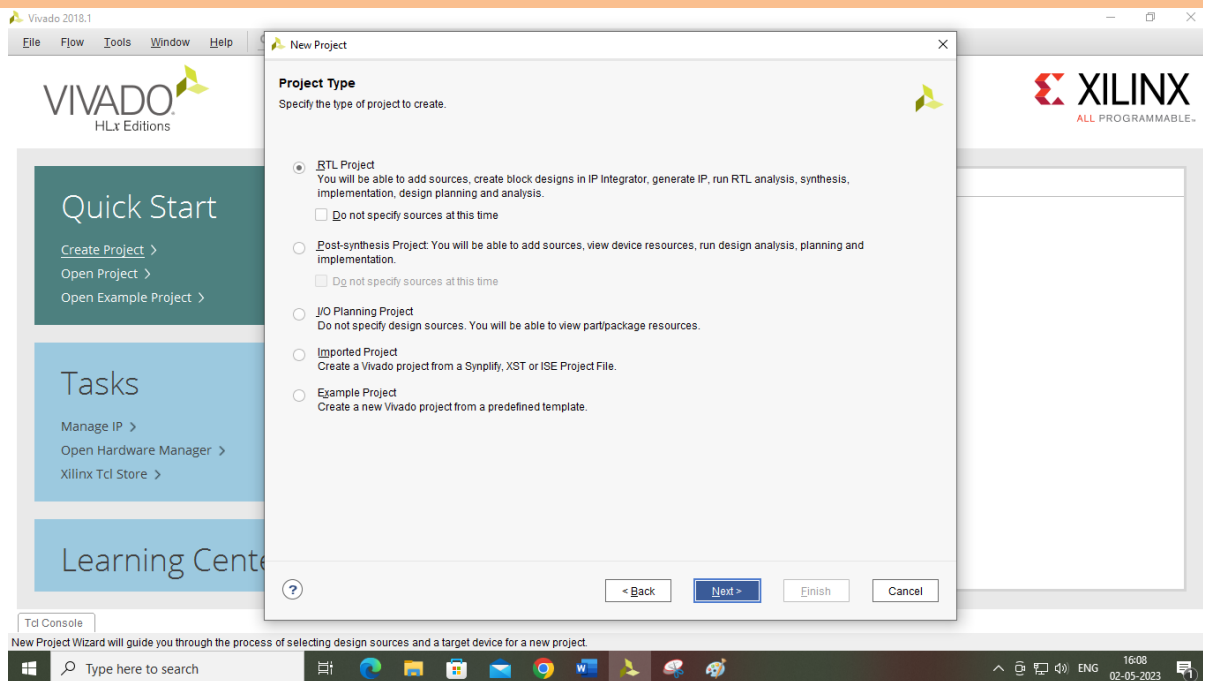
3



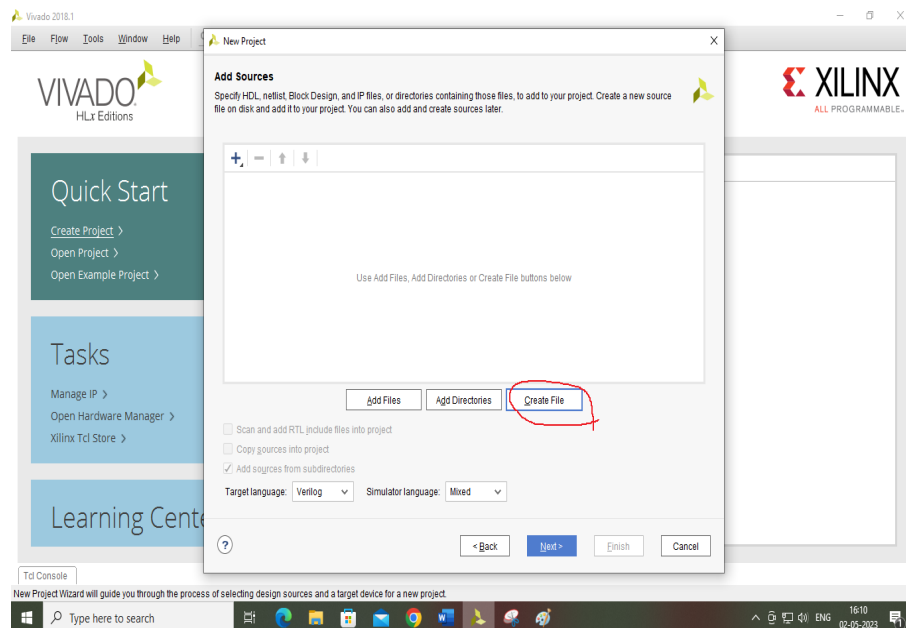
4



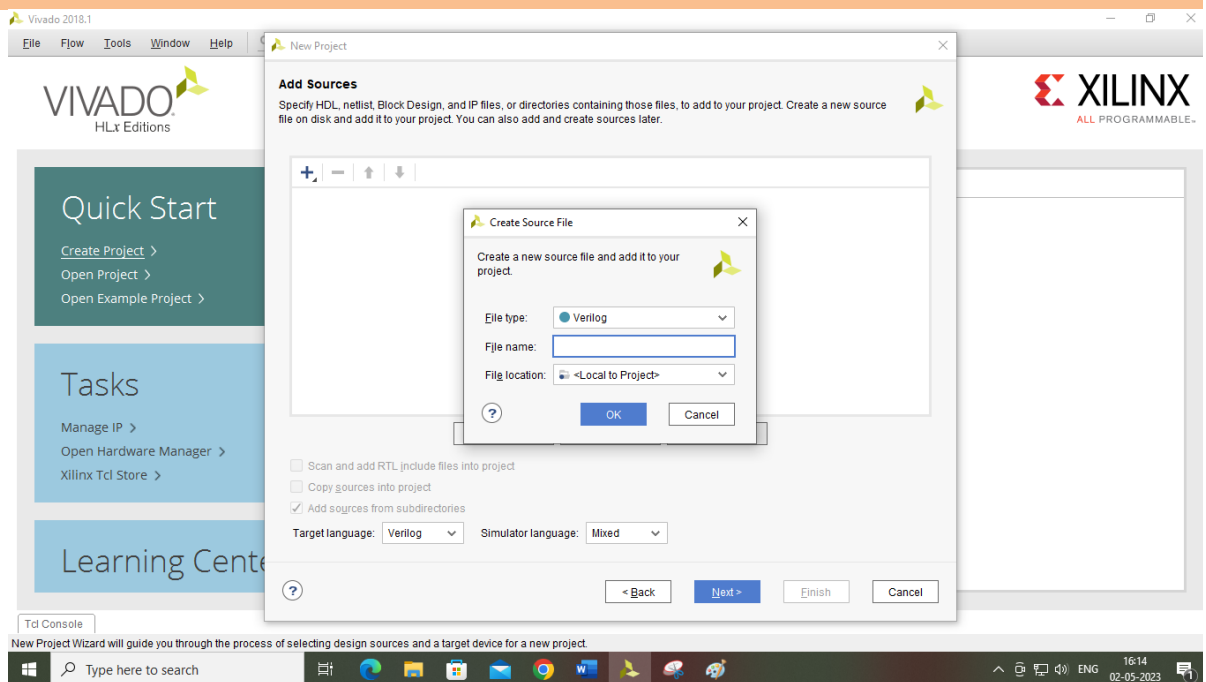
5



6

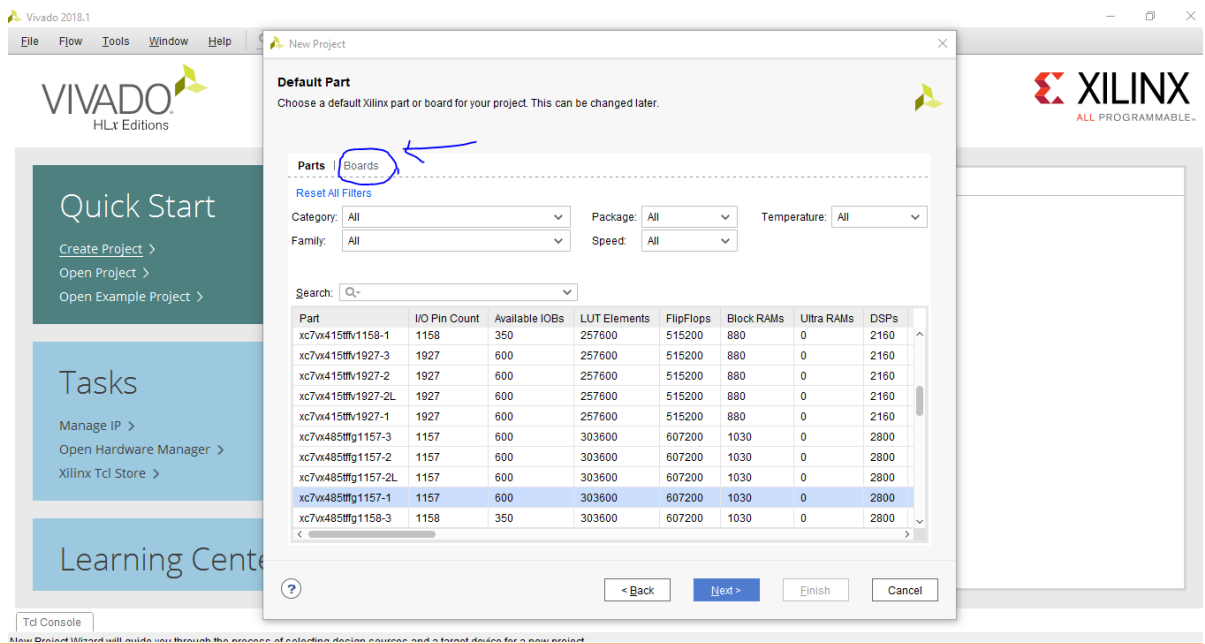


7

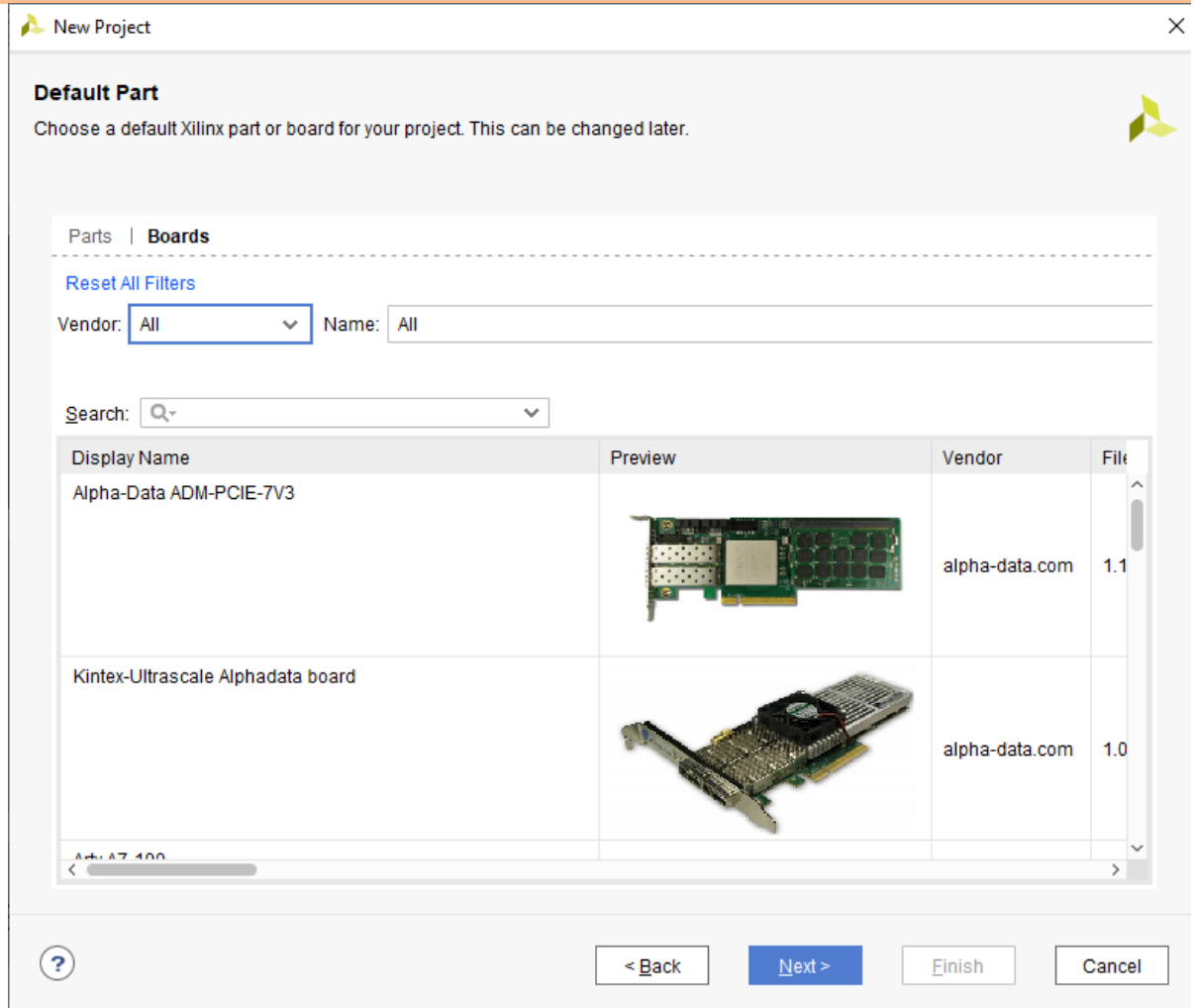


File Type: Verilog
File Name: <myinv1>
File Location: <Local to Project>
OK->NEXT->NEXT

8

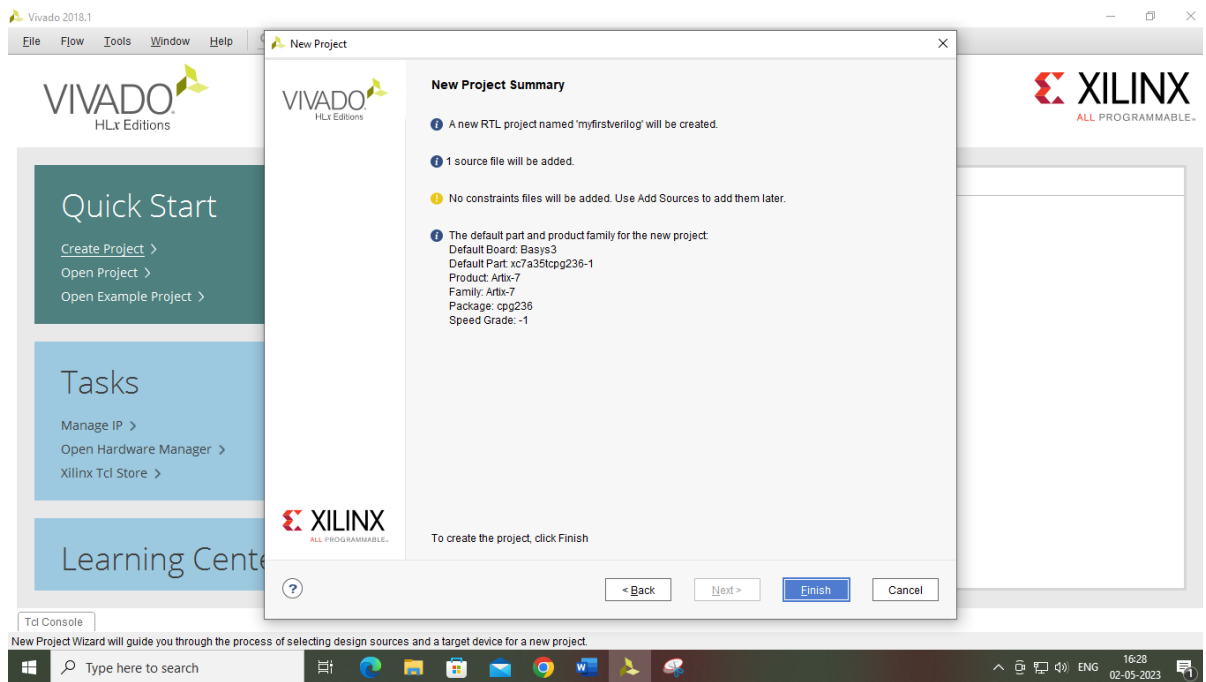


9



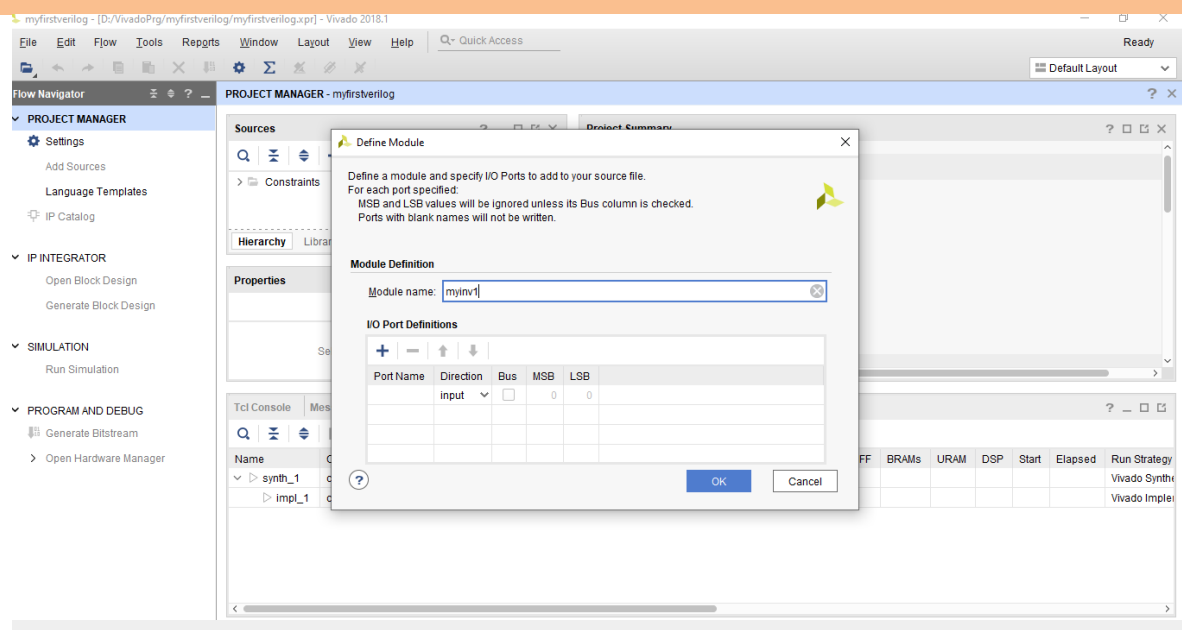
SEARCH BASYS 3-> SELECT BOARD-> NEXT->

10



FINISH

11



DEFINE IO.

12

Define Module

Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

Module name:

I/O Port Definitions

Port Name	Direction	Bus	MSB	LSB
SW1	input	☐	0	0
LED1	output	☐	0	0

13

myfirstverilog - [D:/VivadoPrg/myfirstverilog/myfirstverilog.xpr] - Vivado 2018.1

File Edit Flow Tools Reports Window Layout View Help Q- Quick Access Ready

Flow Navigator PROJECT MANAGER - myfirstverilog

Sources

Design Sources (1)

myinv1 (myinv1.v)

Properties

Select an object to see properties

Project Summary

Settings Edit

Project name: myfirstverilog

Project location: D:/VivadoPrg/myfirstverilog

Product family: Artix-7

Project part: Basys3 (xc7a35tqpg236-1)

Top module name: myinv1

Target language: Verilog

Simulator language: Mixed

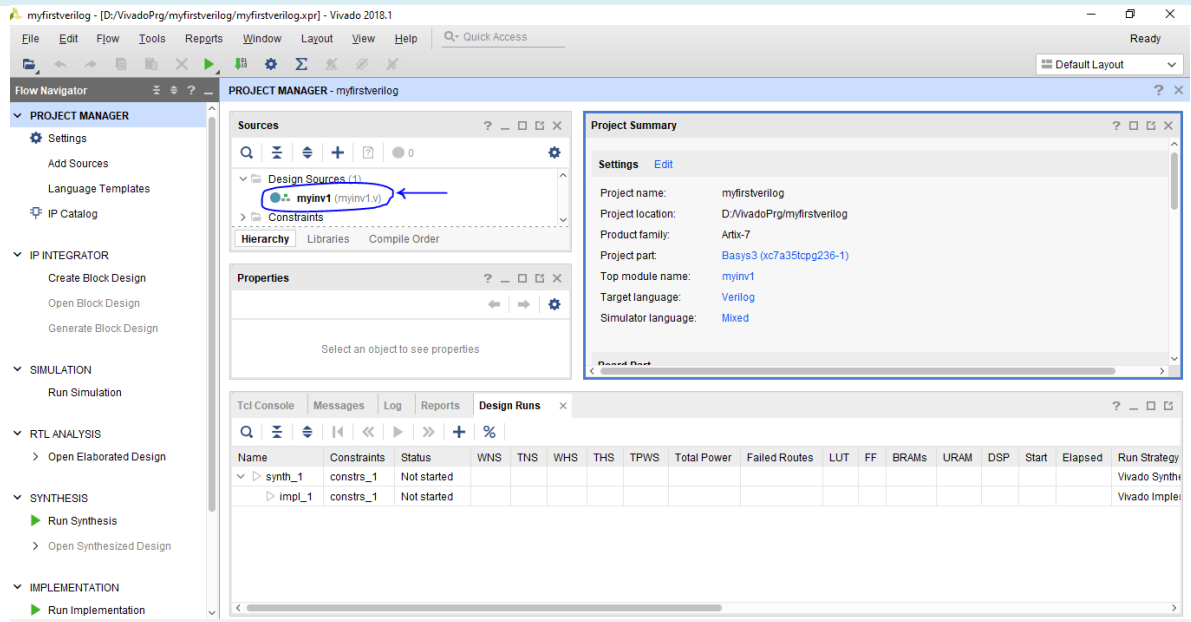
Design Runs

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMS	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started															Vivado Synthesis
impl_1	constrs_1	Not started															Vivado Implementation

STEPS

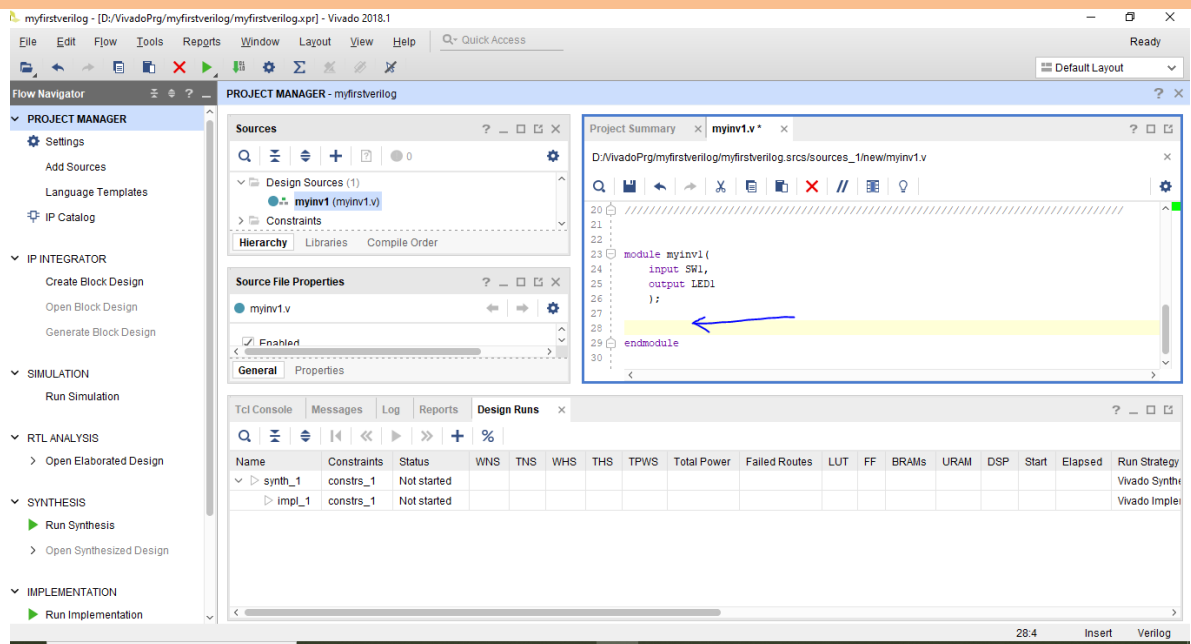
2. PROGRAM & SCHEMATIC

1



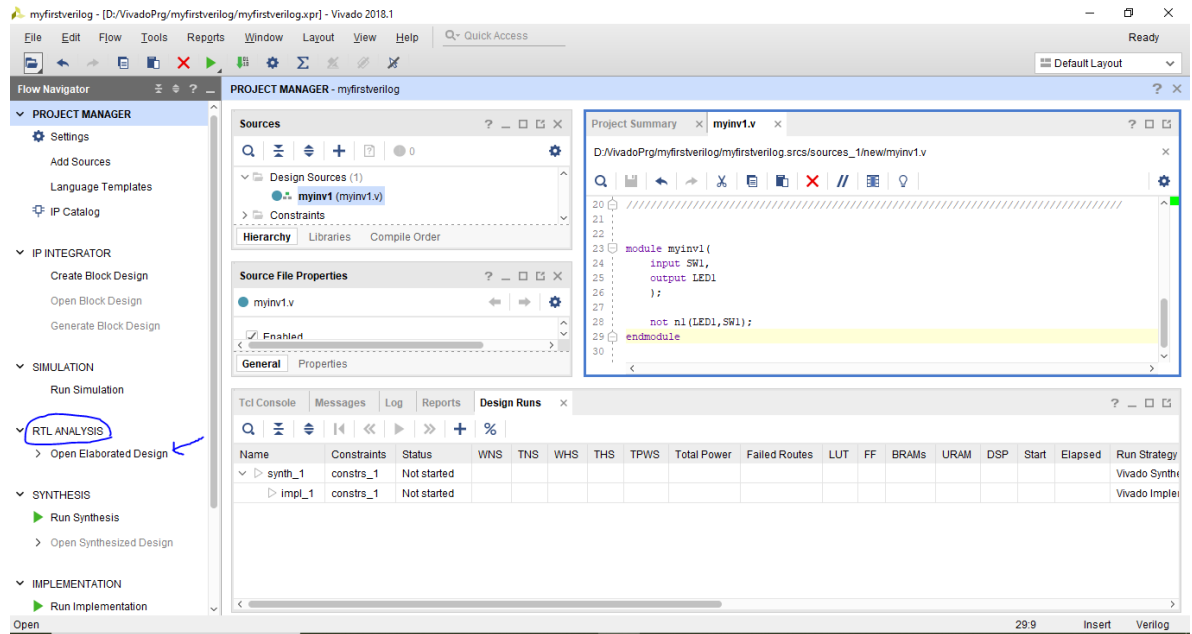
DOUBLE CLICK

2



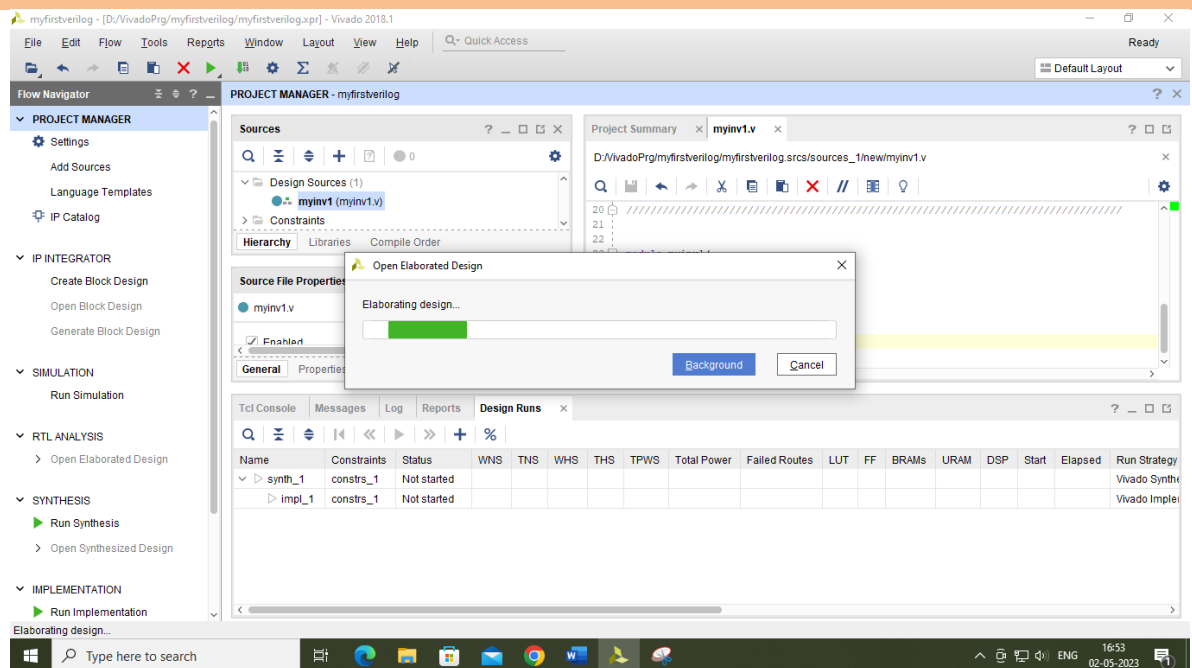
ENTER VERILOG CODE
not n1(LED1,SW1);
SAVE PROGRAM

4



CLICK OPEN ELABORATED DESIGN

5



6

ELABORATED DESIGN - xc7a35tcbg236-1 (active)

Sources Netlist x ? _ □ ✕

myinv1
 > Nets (2)
 > Leaf Cells (1)

Source File Properties ? _ □ ✕

myinv1.v ◀ ▶ ⚙

General Properties

Project Summary x Schematic x myinv1.v x ? _ □ ✕

1 Cell 2 I/O Ports 2 Nets

Tcl Console Messages Log Reports Design Runs x ? _ □ ✕

Q Z ⚙ ⏪ ⏩ ⏴ ⏵ + %

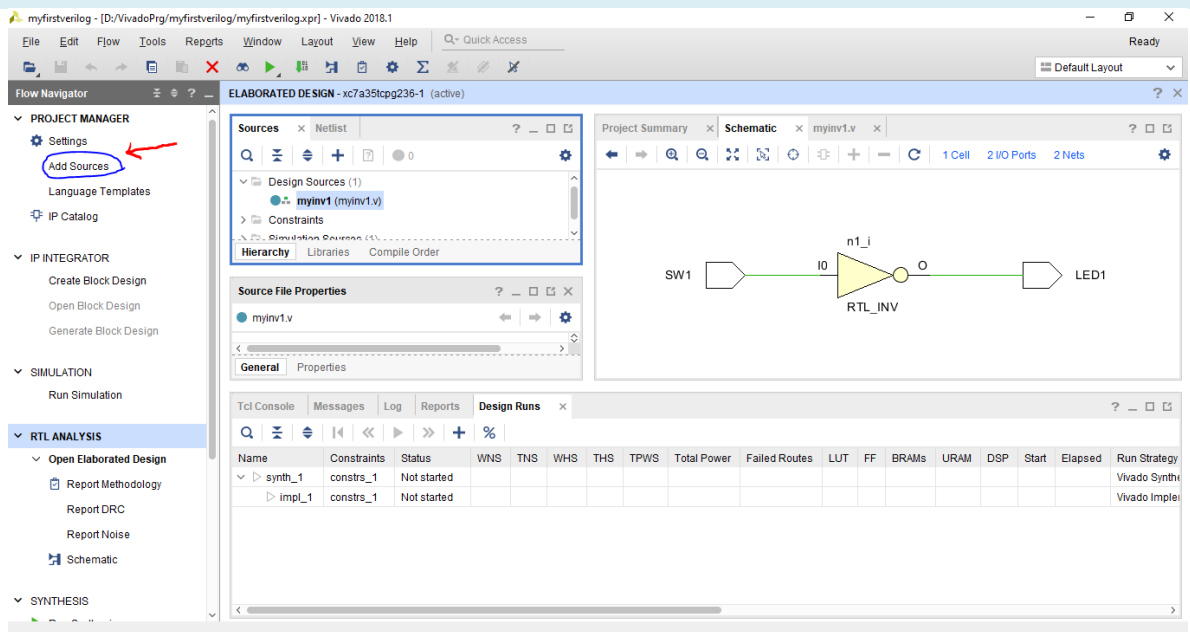
Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAMs	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started															Vivado Synthe
impl_1	constrs_1	Not started															Vivado Implei

CHECK SCHEMATIC DIAGRAM

STEPS

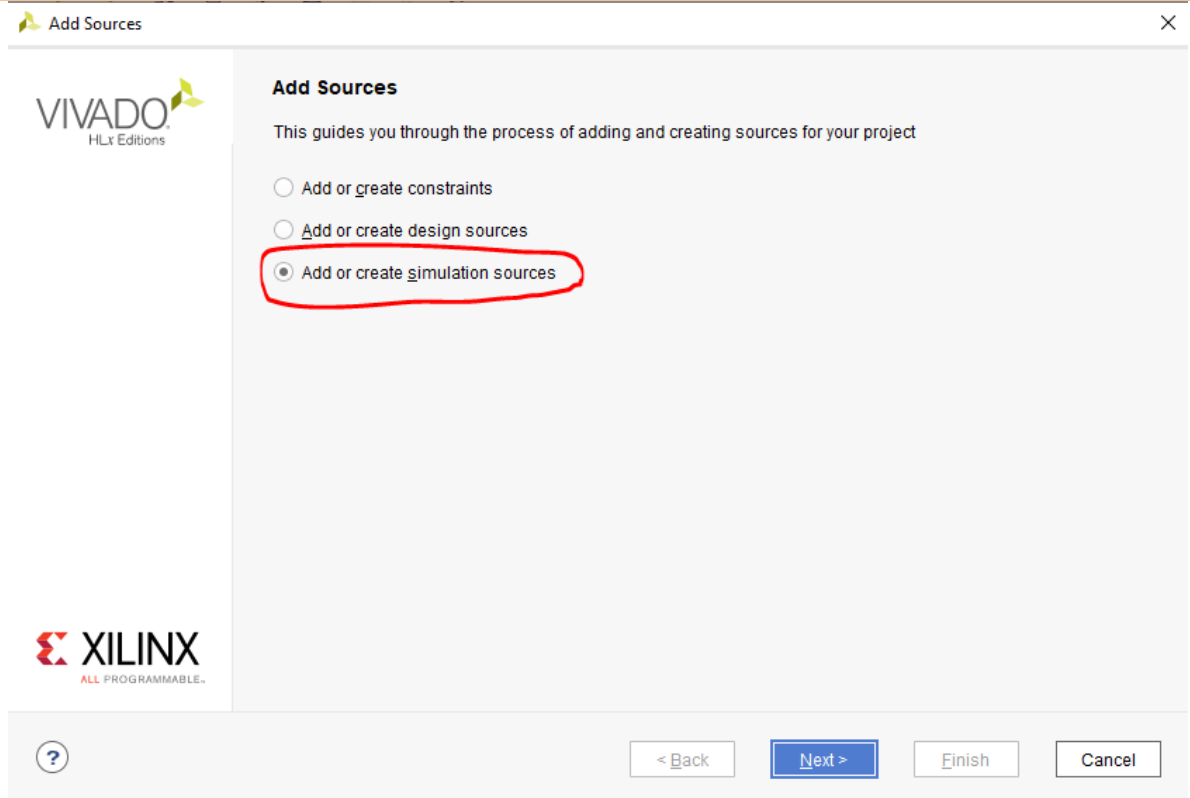
3. GENERATING TESTBENCH

1



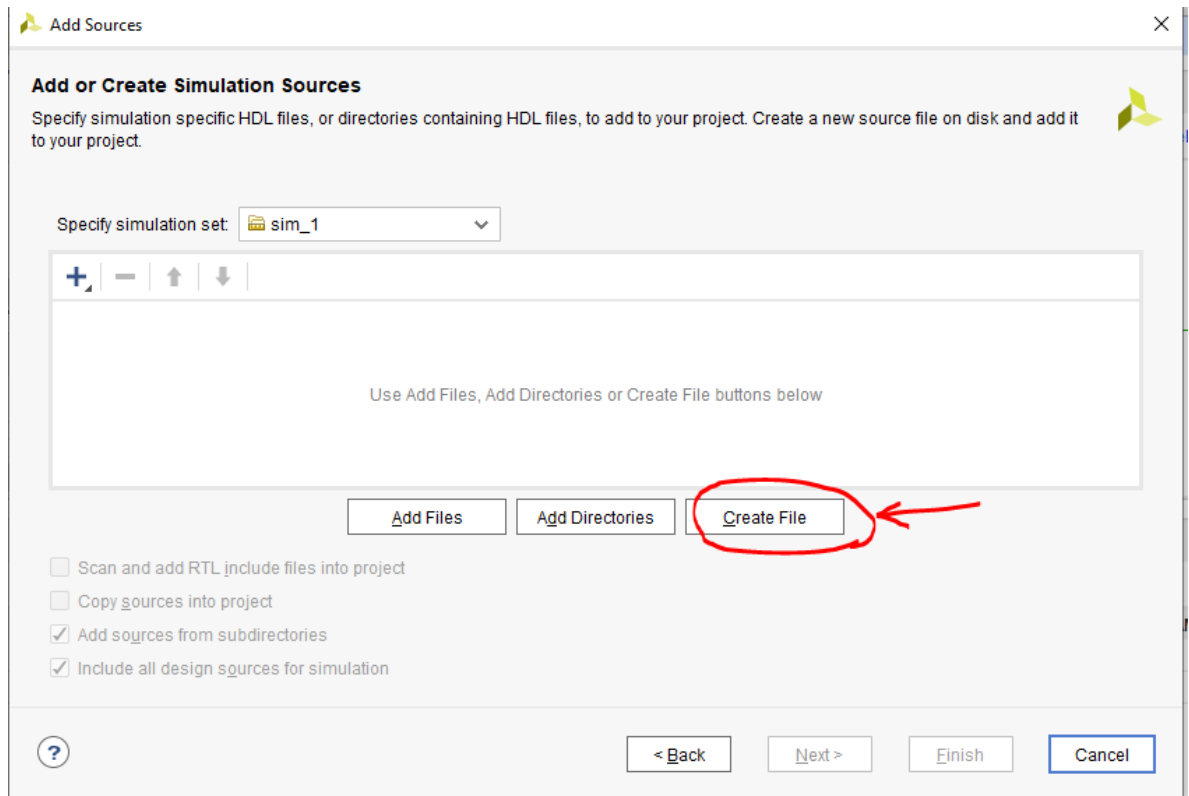
CLICK ADD SOURCES

2



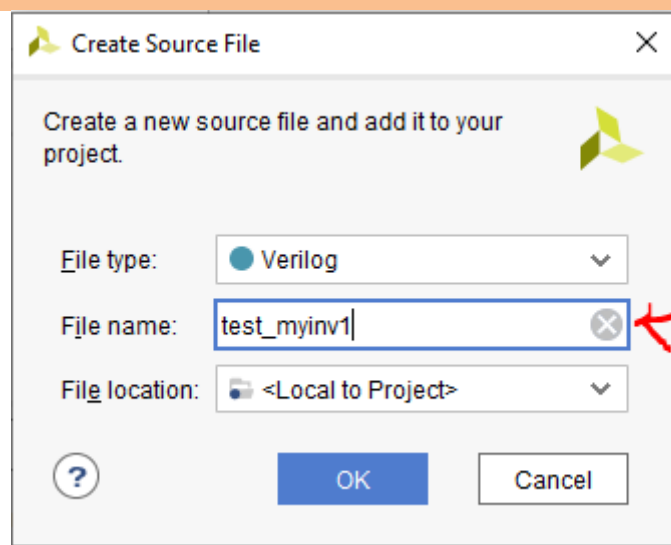
NEXT

3



CREATE FILE

4



File Type: Verilog
File Name: <test_myinv1>
File Location: <Local to Project>
OK

5

Add Sources

Add or Create Simulation Sources

Specify simulation specific HDL files, or directories containing HDL files, to add to your project. Create a new source file on disk and add it to your project.

Specify simulation set: sim_1

	Index	Name	Library	Location
●	1	test_myinv1.v	xil_defaultlib	<Local to Project>

☐ Scan and add RTL include files into project
☐ Copy sources into project
☒ Add sources from subdirectories
☒ Include all design sources for simulation

6

Define Module

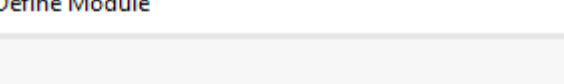
Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

Module name: test_myinv1

I/O Port Definitions

Port Name	Direction	Bus	MSB	LSB
	input	☐	0	0



Define Module

?

The module definition has not been changed.
Are you sure you want to use these values?

Yes No

Sources ? _ □ ↗ ✕

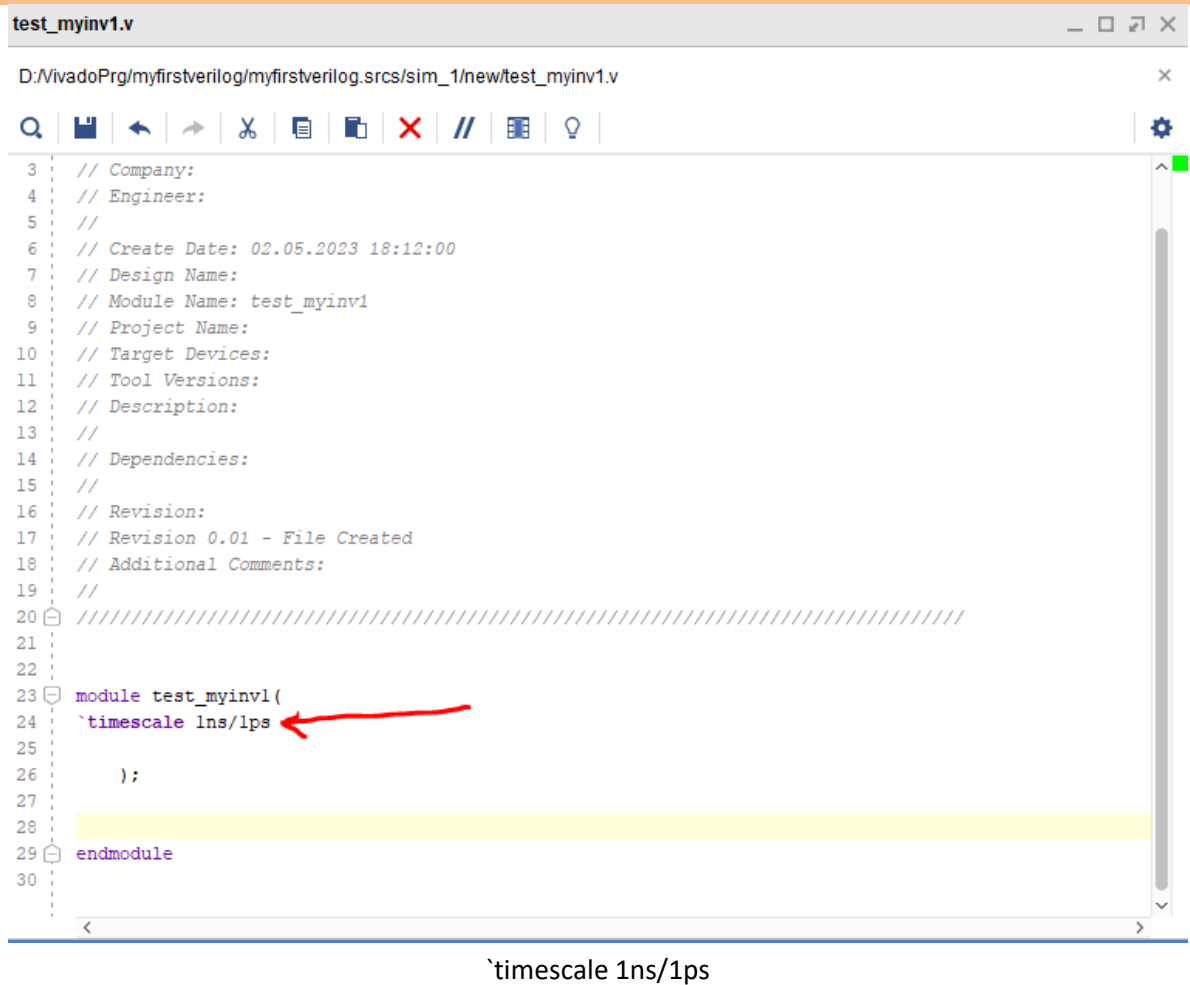
🔍 ⚡ ⚖️ + ? 0 ⚙️

- ▼ 📁 Design Sources (1)
 - 📄 **myinv1** (myinv1.v)
- > 📁 Constraints
- ▼ 📁 Simulation Sources (2)
 - ▼ 📁 sim_1 (2)
 - 📄 **myinv1** (myinv1.v)
 - 📄 **test_myinv1** (test_myinv1.v)

Hierarchy Libraries Compile Order

[illegible]

10



```
test_myinv1.v
D:/VivadoPrg/myfirstverilog/myfirstverilog.srscs/sim_1/new/test_myinv1.v

3 // Company:
4 // Engineer:
5 //
6 // Create Date: 02.05.2023 18:12:00
7 // Design Name:
8 // Module Name: test_myinv1
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module test_myinv1(
24     `timescale 1ns/1ps
25
26     );
27
28
29 endmodule
30
```

``timescale 1ns/1ps`

11

```

test_myinv1.v
D:/VivadoPrg/myfirstverilog/myfirstverilog.srscs/sim_1/new/test_myinv1.v

14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module test_myinv1(
24     `timescale 1ns/1ps
25
26 );
27
28     reg SW1;
29     wire LED1;
30
31     myinv1 CUT (SW1,LED1);
32     initial
33     begin
34
35         SW1=0; #100;
36         SW1=1; #100;
37         SW1=0; #100;
38
39     end
40
41 endmodule
42

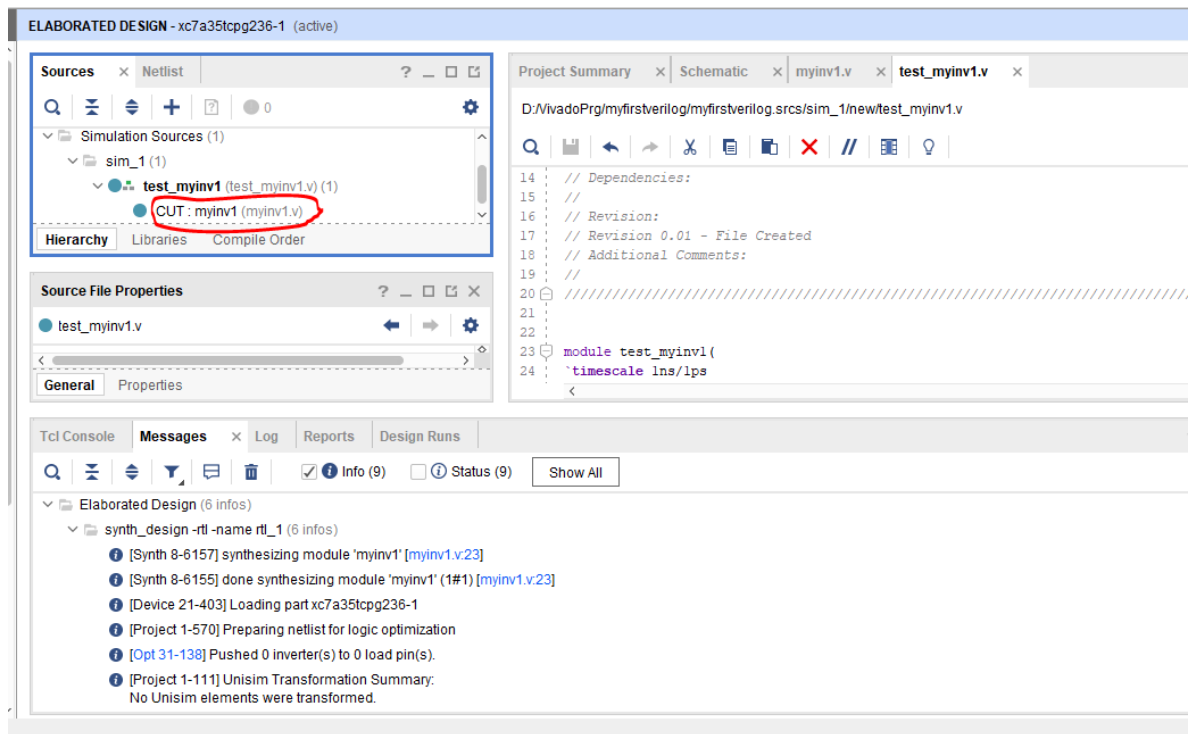
```

```

myinv1 CUT (SW1,LED1);
    initial
    begin
        SW1=0; #100;
        SW1=1; #100;
        SW1=0; #100;
    end
SAVE PROGRAM

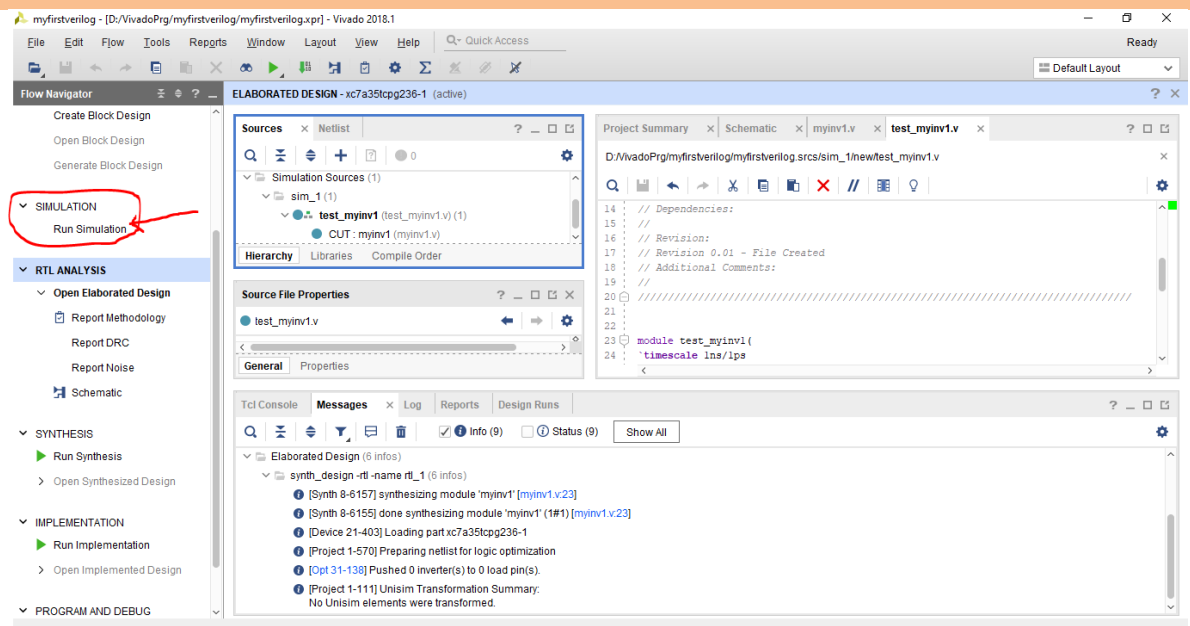
```


12



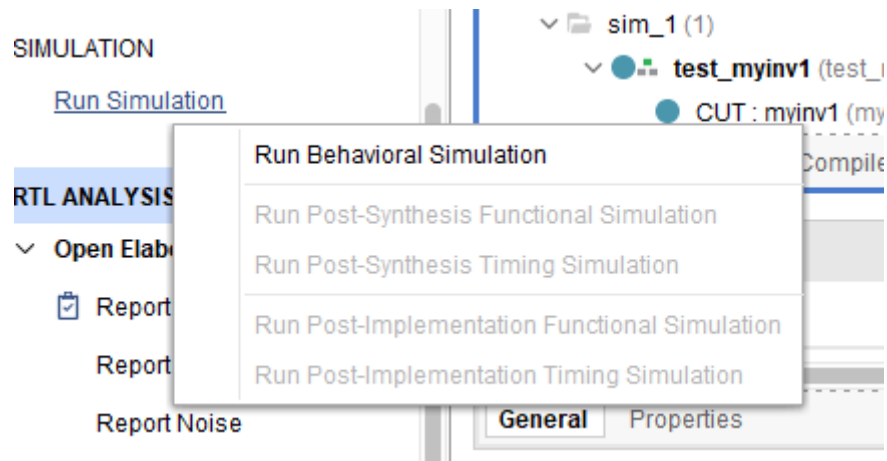
CUT FILE CREATED

13



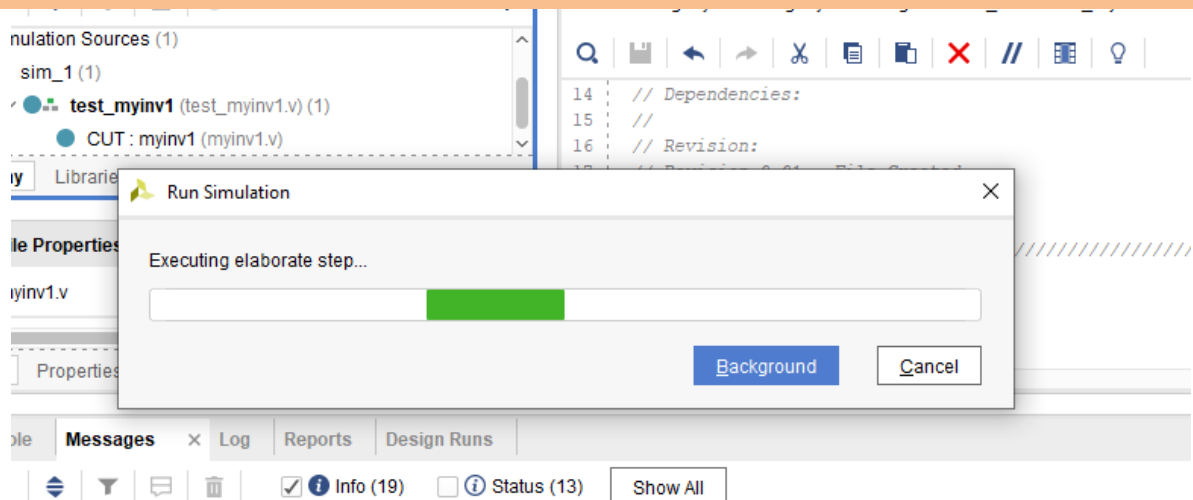
CLICK

14

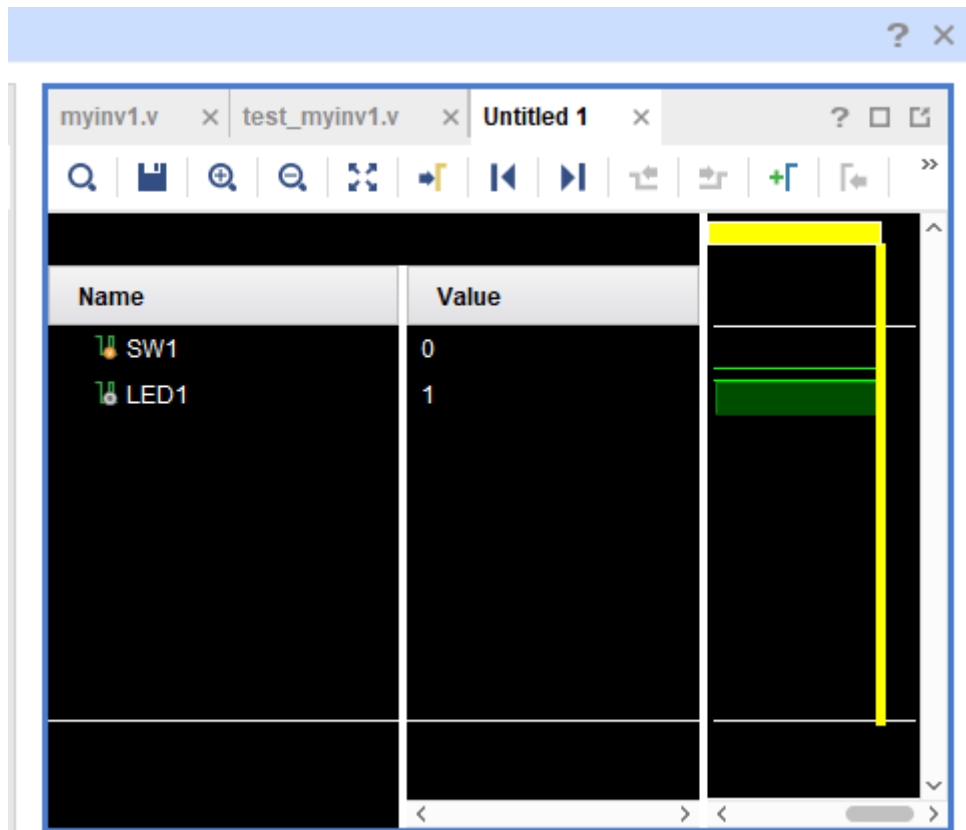


CLICK-> RUN BEHAVIORAL SIMULATION

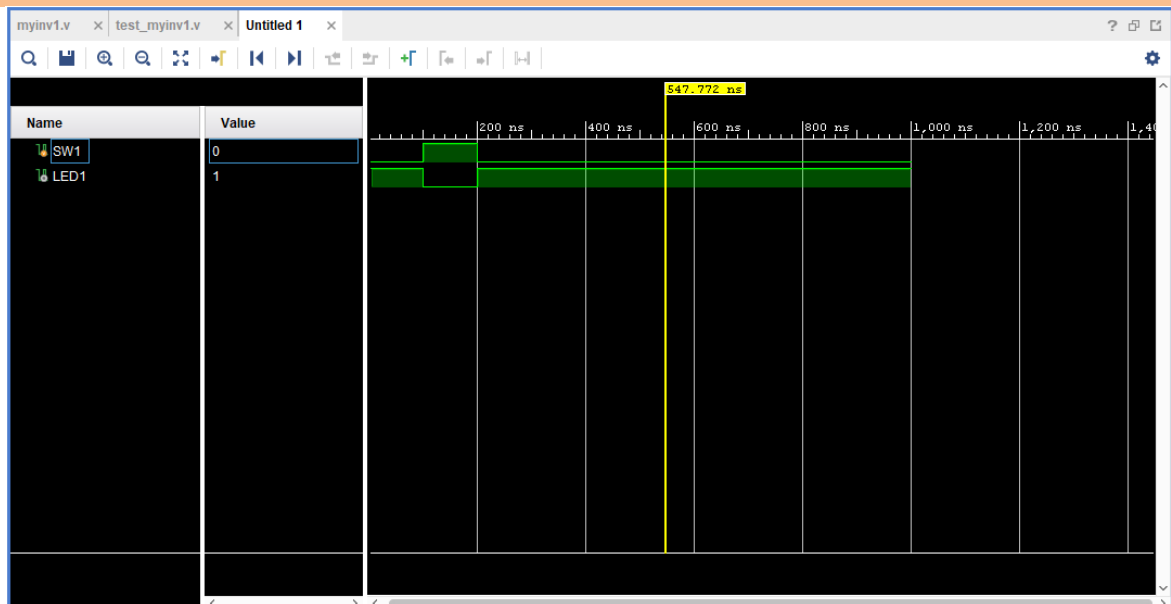
15



16



17

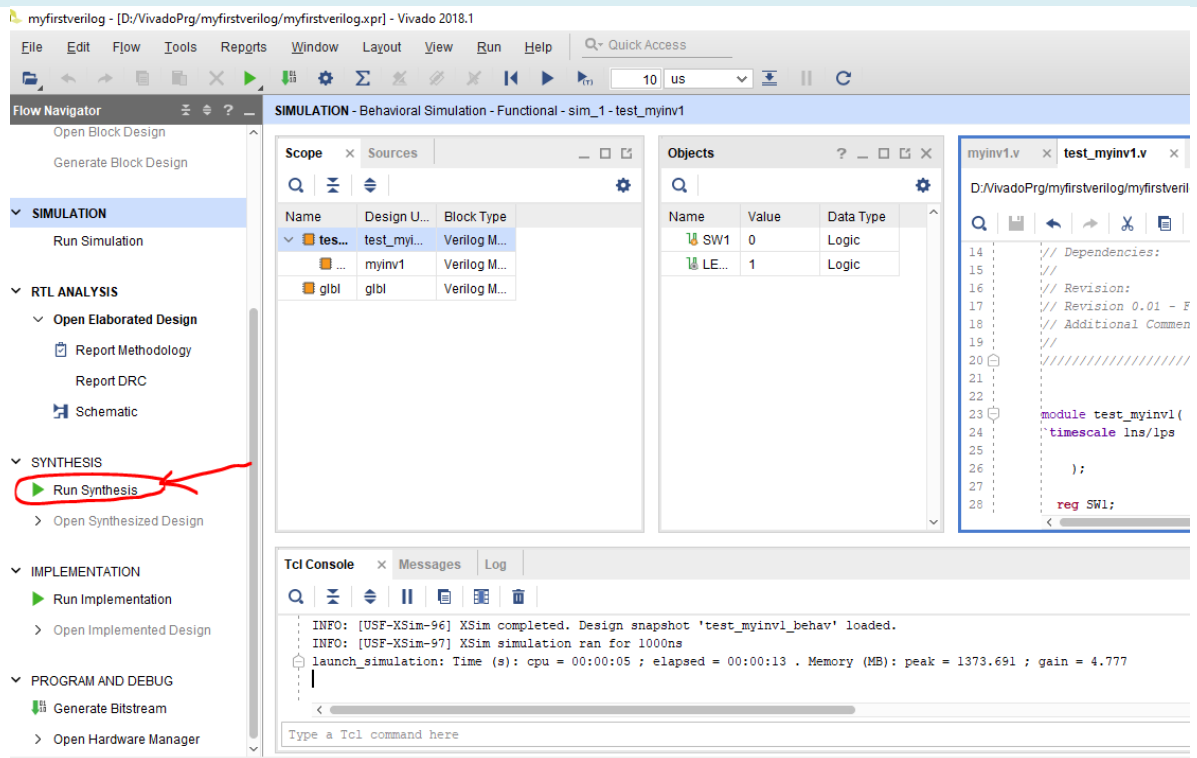


CHECK THE WAVEFORM

STEPS

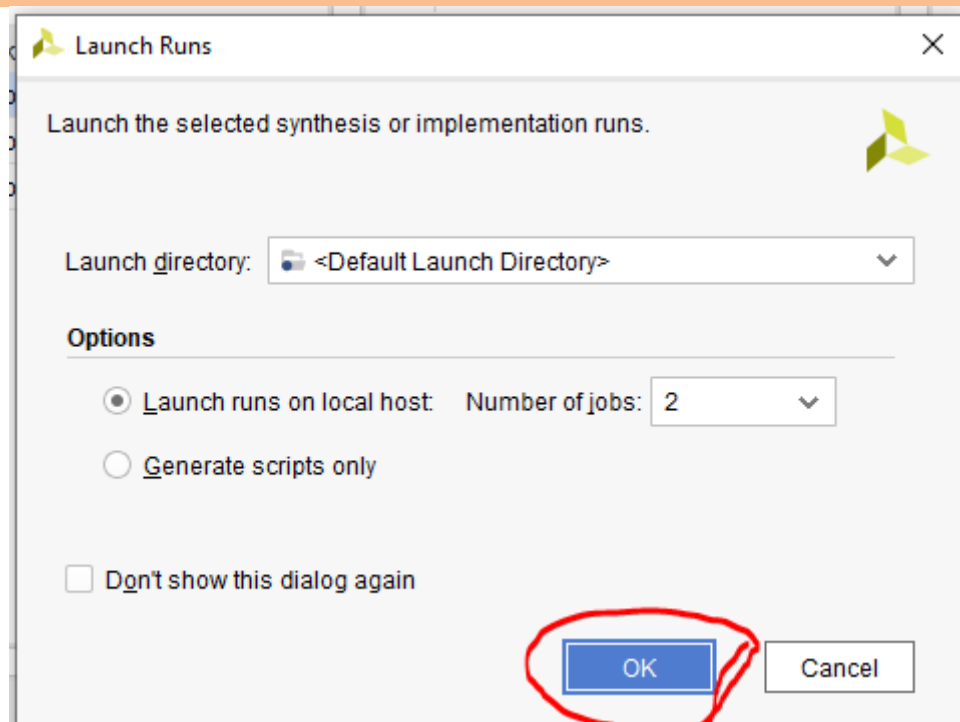
4. SYNTHESIS

1



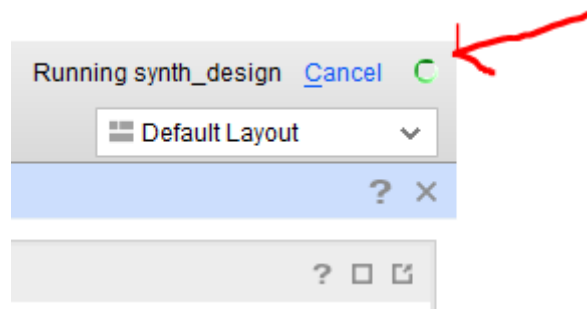
CLICK

2



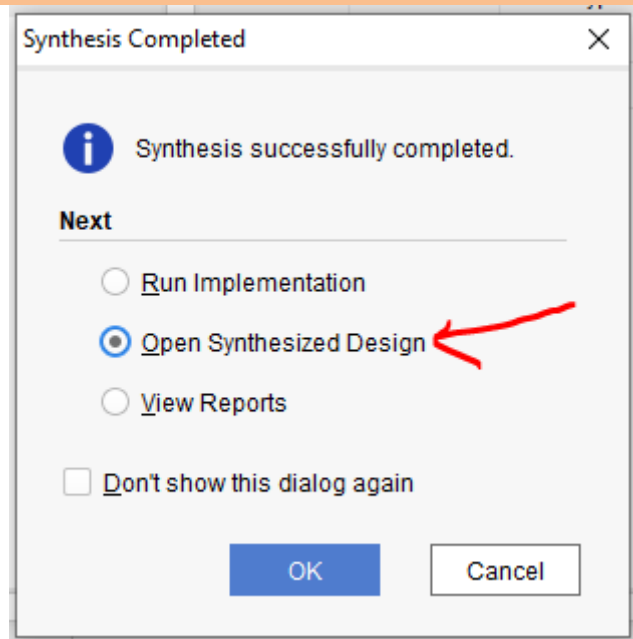
CLICK OK

3



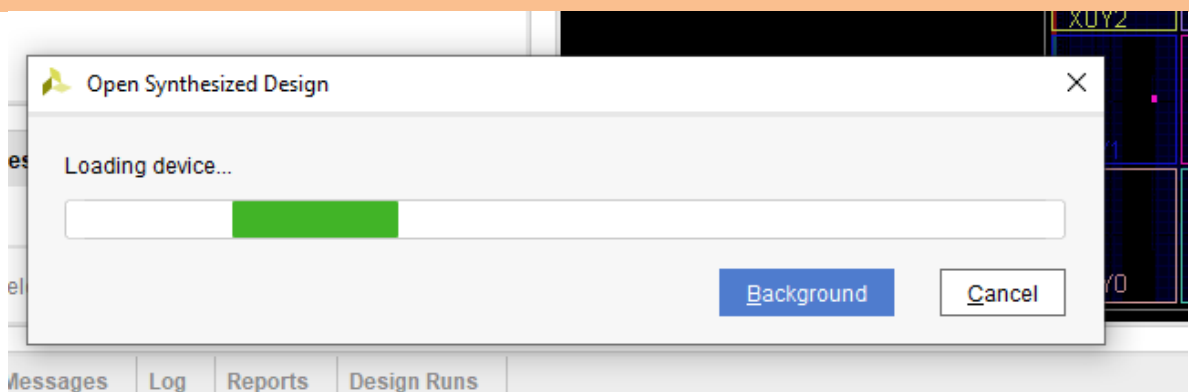
WAIT TILL COMPLETION OF THE PROCESS

4

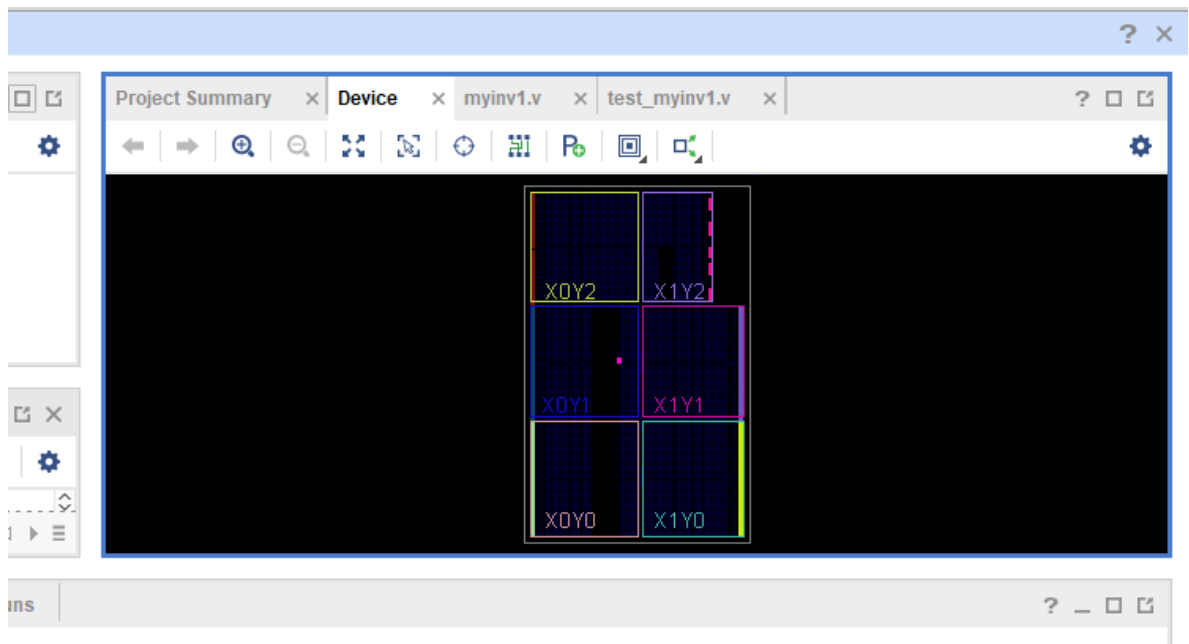


OK

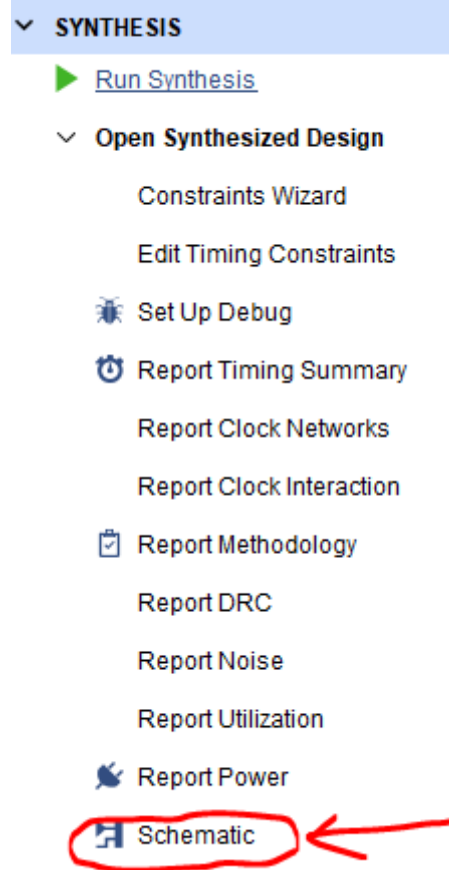
5



6

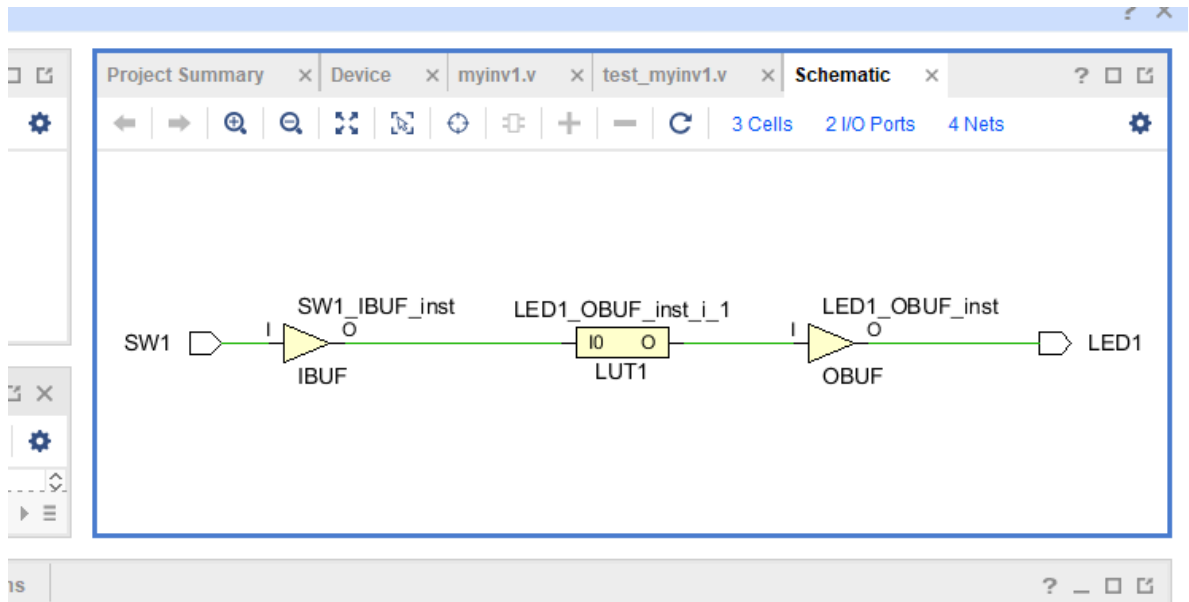


7

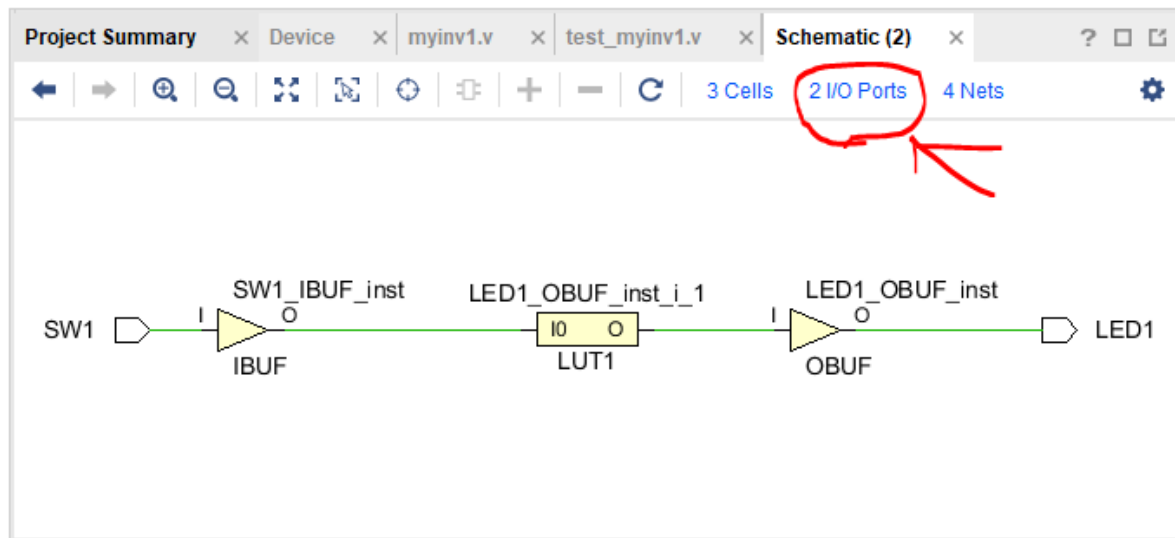


CLICK

8



9



CLICK

10

I/O Port Properties

Name: SW1

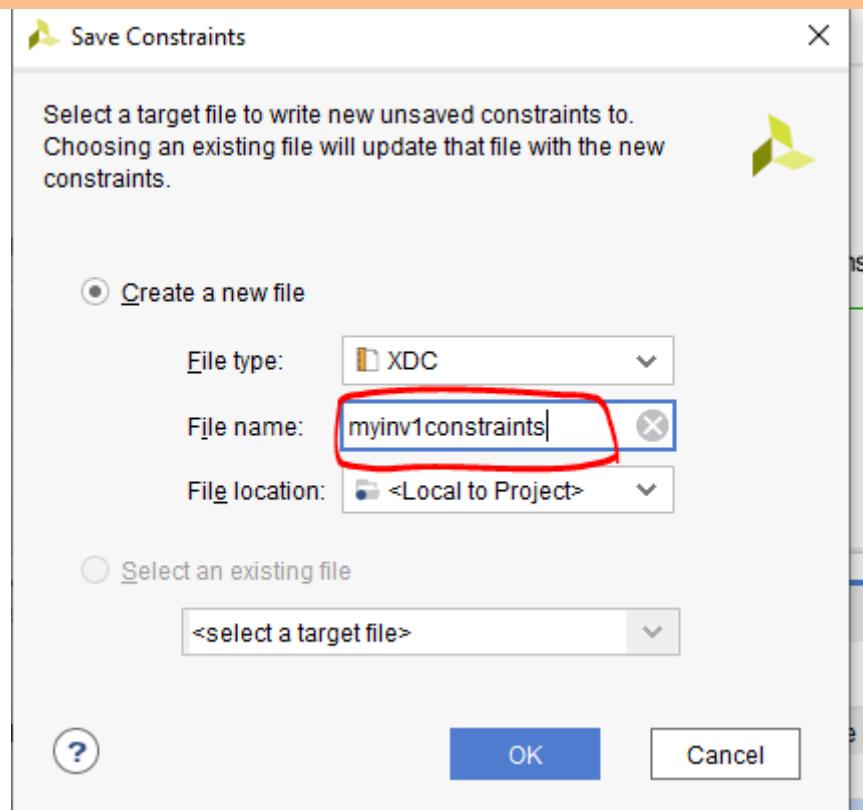
General Properties Configure Power

Name	Direction	Board Part Pin	Board Part Interface	Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive
LED1	OUT					L1	☒	35	LVCMOS33*	3.300		12
SW1	IN					V17	☒	14	LVCMOS33*	3.300		

I/O Ports in 'Schematic (2)' (2)

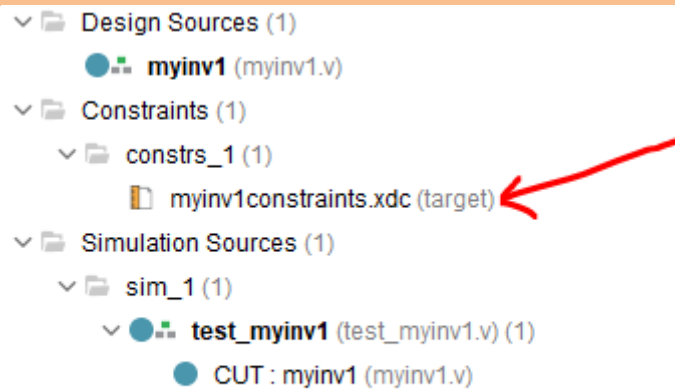
SAVE-> OK

11

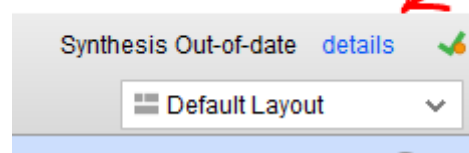


OK

12

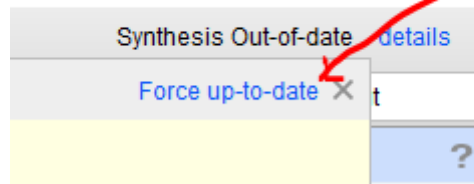


13



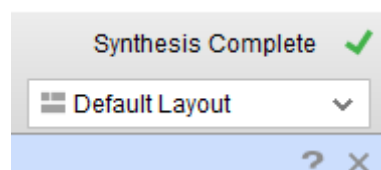
CLICK DETAILS

14



CLICK

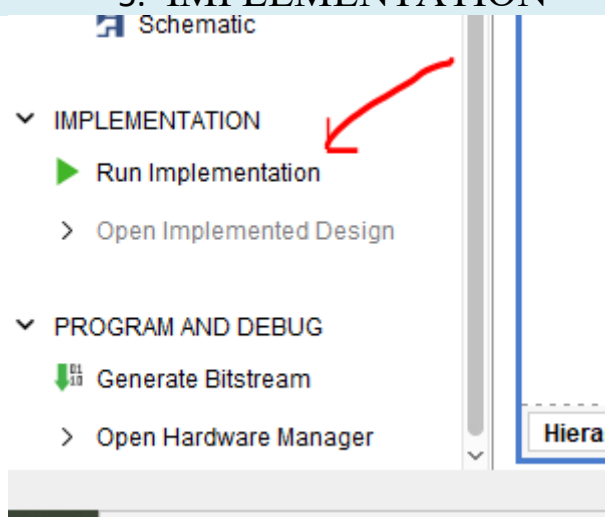
15



STEPS

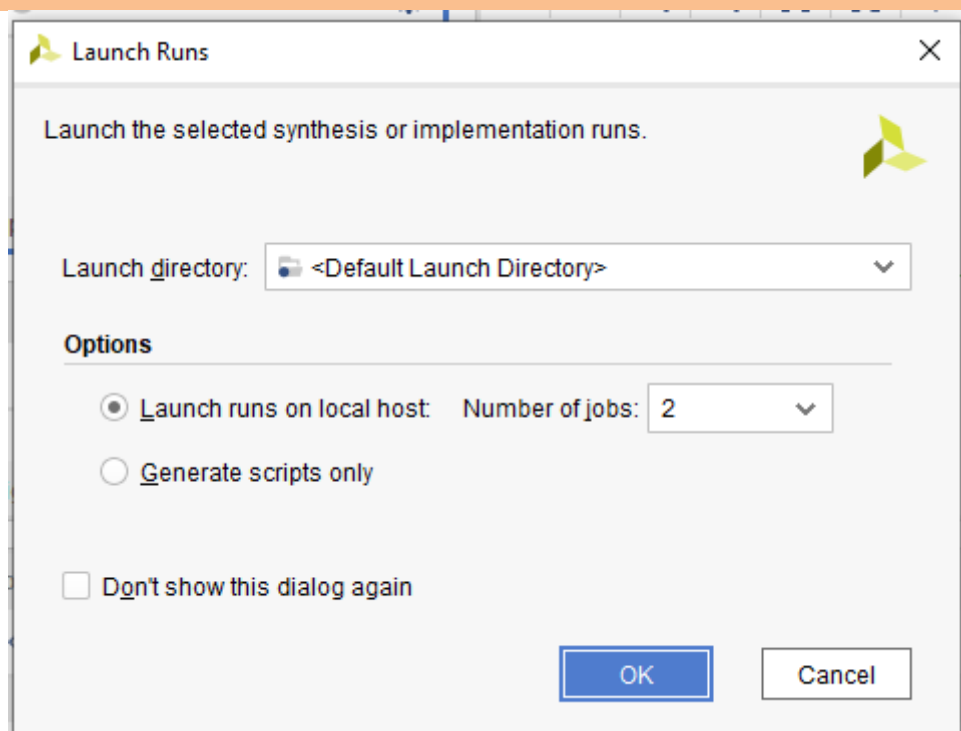
5. IMPLEMENTATION

1



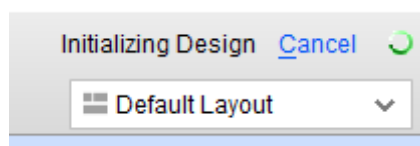
CLICK

2



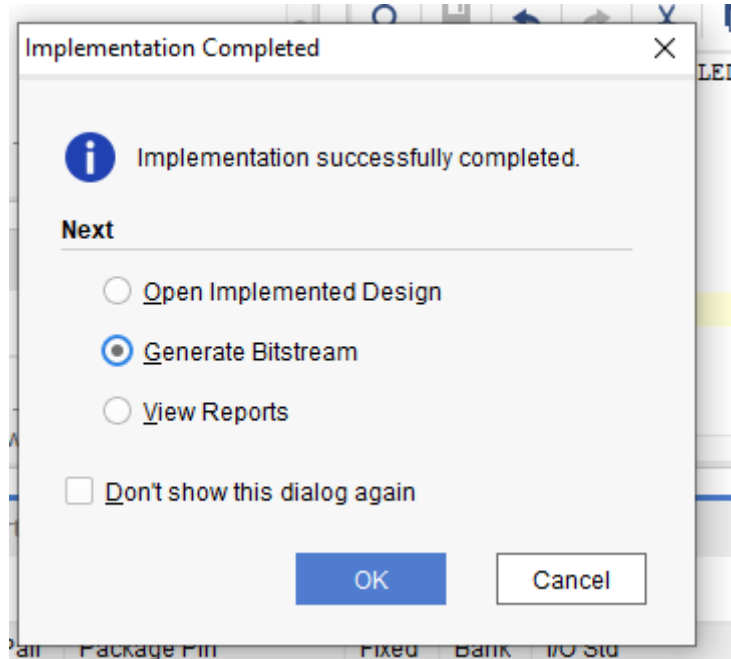
OK

3



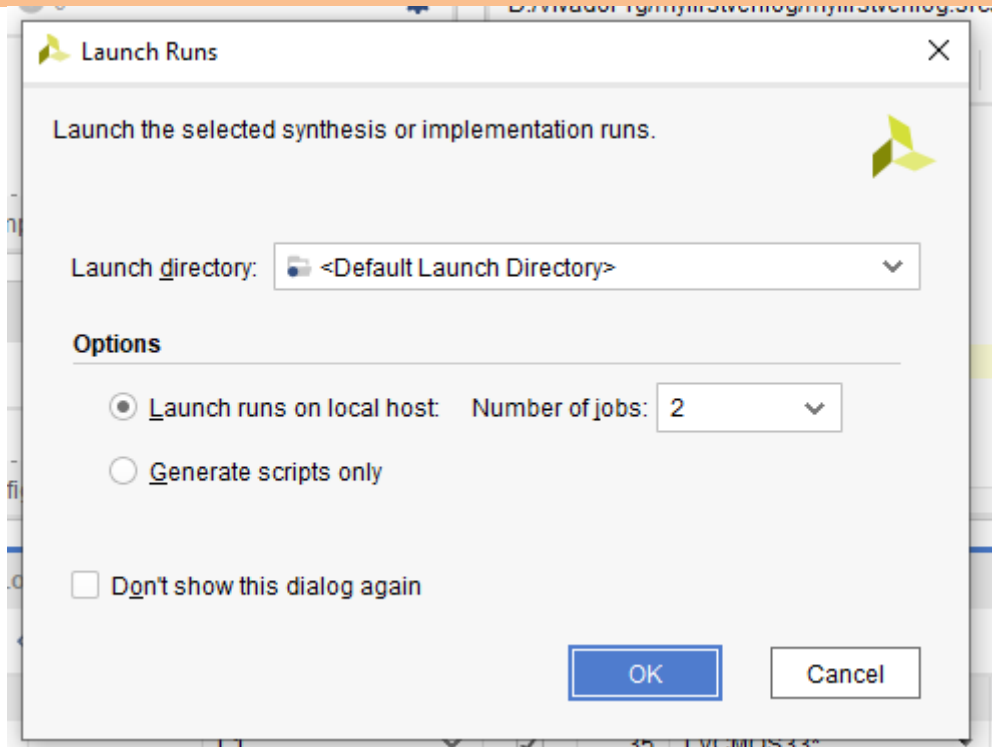
CHECK STATUS

4



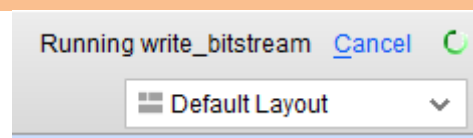
OK

5



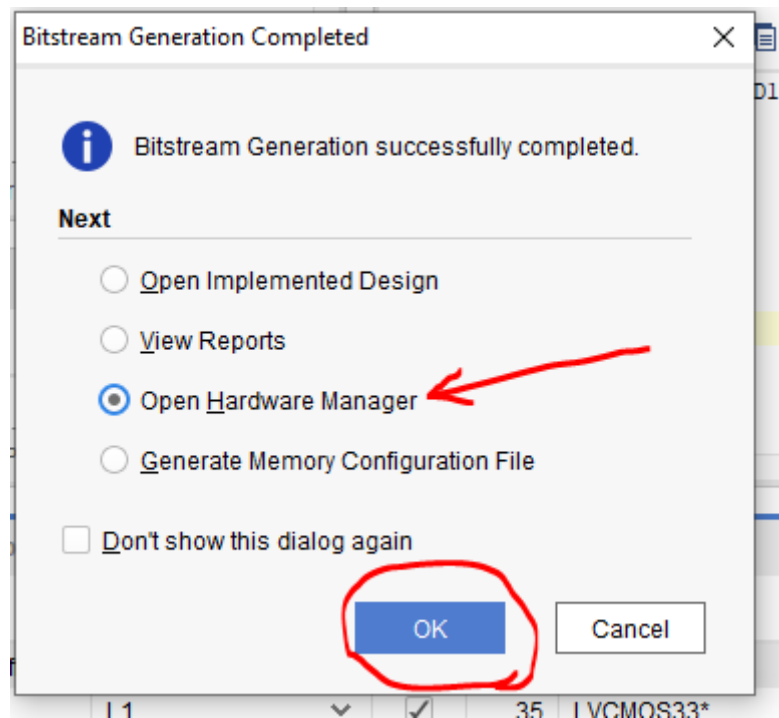
OK

6



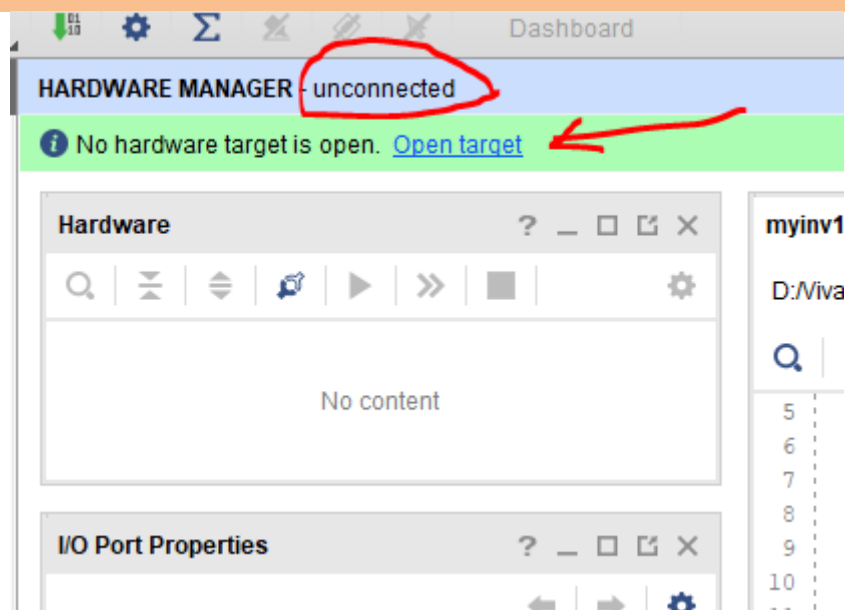
CHECK STATUS

7



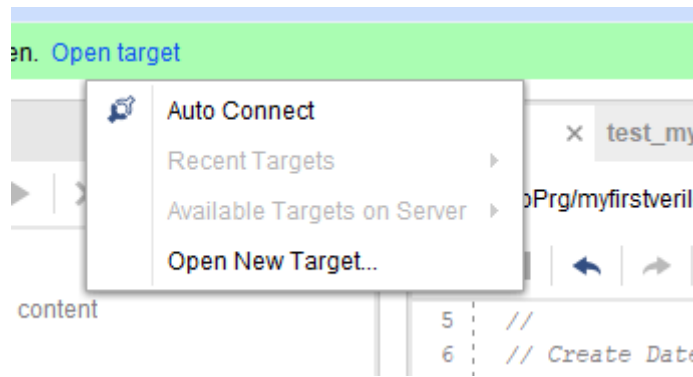
OK

8



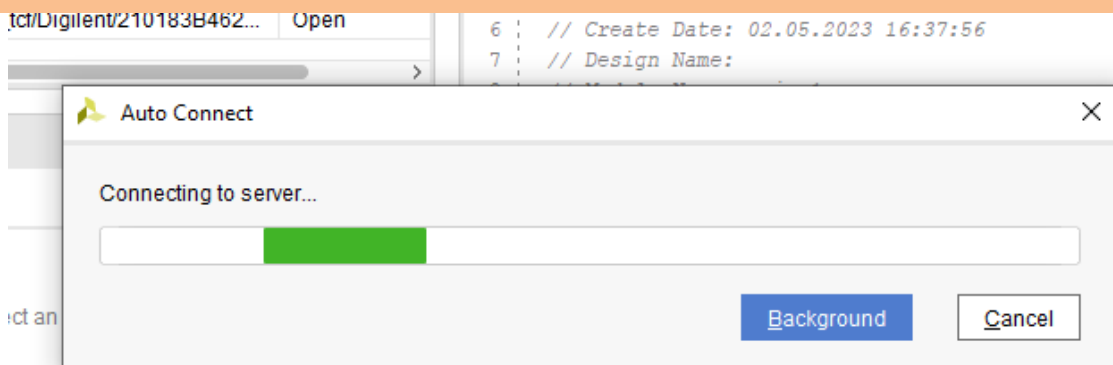
CONNECT DEVICE

9

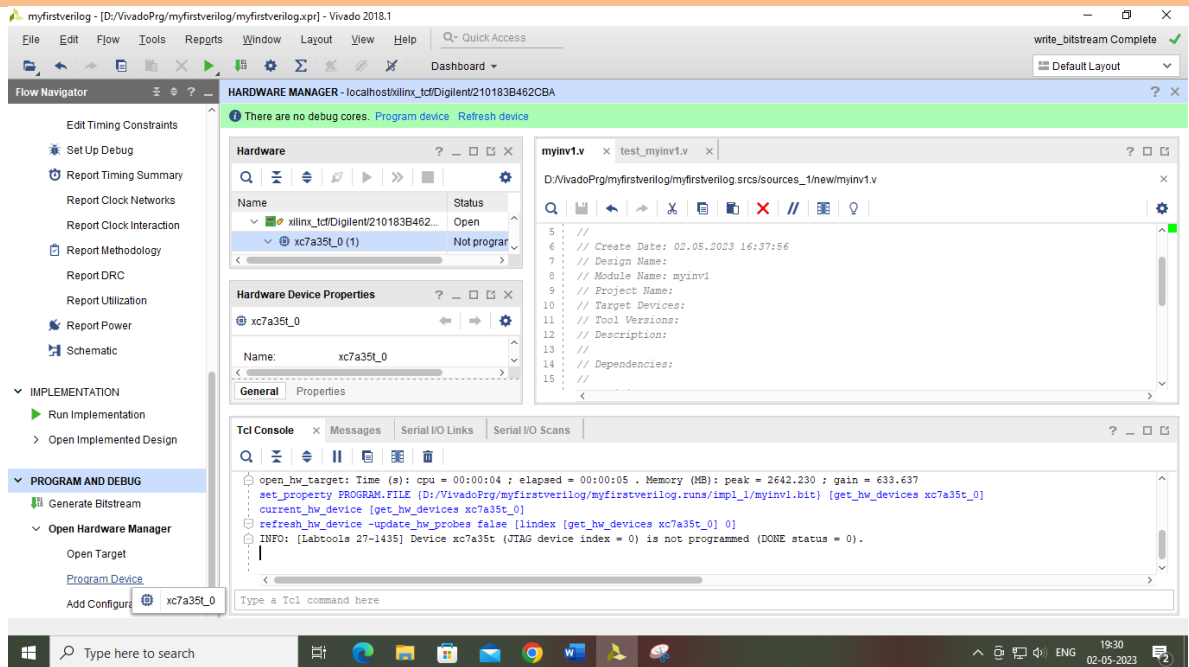


OPEN TARGET-> AUTO CONNECT

10

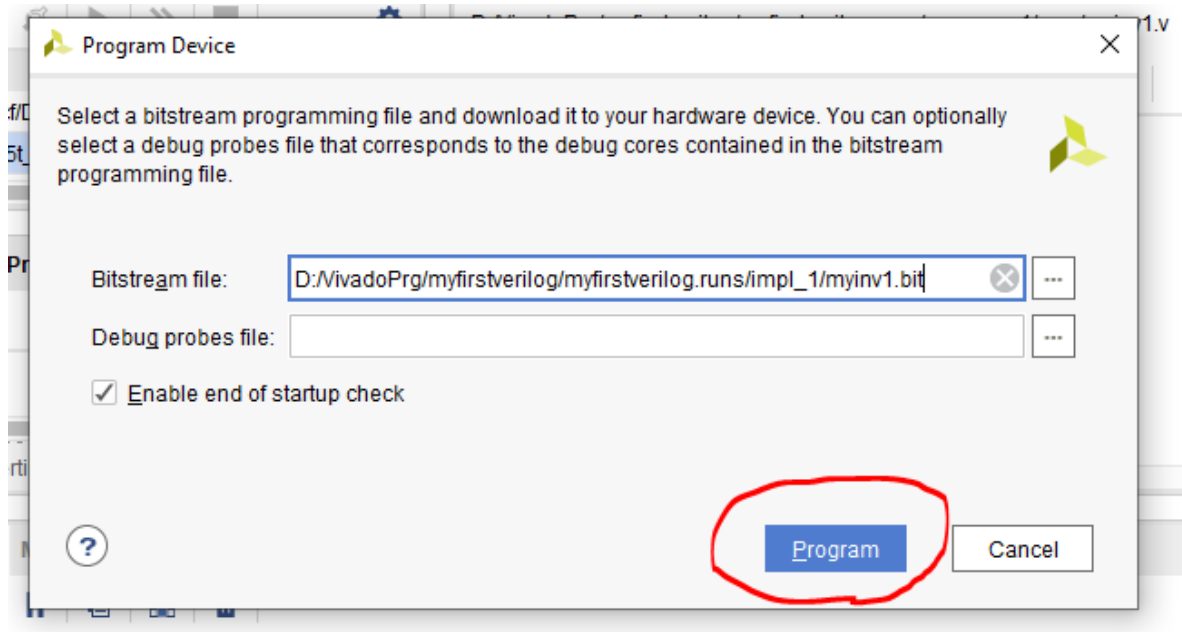


11



CLICK PROGRAM DEVICE-> xc7a35t_0

12



PROGRAM

13

CHANGE INPUT AND CHECK THE OUTPUT